# Quiz #3 (Unit 3 - Extended): ARM Assembly & RISC Pipeline Architecture

Total Marks: 24.00

Instruction Decoding, Register Execution, Pipeline Hazards & Datapaths

**Name:** **Student ID:** **Score:** / 24.00

> **Instructions:** Read the technique hint preceding each question. Concept questions are formatted as *[Multiple Choice]* (single correct answer) or *[Select All That Apply]* (one or more correct answers). Calculation and machine decoding questions are formatted as *[Fill in the Blank]*. Calculators are not permitted.

## PART 1: ARM ASSEMBLY LANGUAGE & REGISTER ARITHMETIC

### Question 1

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** ARM data movement semantics: loading immediate values to initialize or reset register contents.

What is the primary purpose of executing the ARM instruction `MOV R4, #0`?

- a. To subtract 0 from register R4
- b. To compare the current value in R4 with 0
- c. To clear register R4 (set its stored value to 0)
- d. To branch to memory address 0 if R4 is active

### Question 2

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** Bitwise identity operations: evaluating bitwise XOR properties where any bit pattern XORed with itself evaluates to zero ($X \oplus X = 0$).

In ARM assembly, which of the following instructions clears register R2 to zero without referencing an immediate value constant?

- a. `EOR R2, R2, R2`
- b. `ORR R2, R2, R2`
- c. `AND R2, R2, R2`
- d. `MVN R2, R2`

## Question 3

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** Three-operand ARM addition syntax ($Rd = Rn + Rm$): update destination register using source register contents.

The instruction `ADD R3, R3, R2` is executed. If register R3 initially contains 28 and register R2 contains 14, what decimal value will be stored in R3?

Value in R3: [ ]

## Question 4

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** Sequential register state propagation: evaluating immediate moves and subsequent cumulative register addition.

Consider the execution of the following sequential ARM instructions:

```
MOV R1, #15
ADD R2, R1, #25
```

What decimal value is stored in register R2 upon completion?

Value in R2: [ ]

## Question 5

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** Immediate subtraction syntax ($Rd = Rn - \text{Imm}$): updating accumulator register in-place.

What is the decimal value in register R6 after the instruction `SUB R6, R6, #18` is executed, assuming R6 initially contained 45?

Value in R6: [ ]

## Question 6

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** ARM Reverse Subtract instruction (RSB): computing destination as $Rd = \text{Operand2} - Rn$.

In ARM assembly, the instruction `RSB R0, R2, #40` performs reverse subtraction ($R0 = 40 - R2$). If register R2 contains 12, what decimal value is stored in R0?

Value in R0: [ ]

## Question 7

Mark 1.00 out of 1.00

[FILL IN THE BLANK - MACHINE DECODING]

**Technique / Concept Hint:** ARM immediate data processing decoding: Rd field located at bits 15-12; 8-bit unsigned immediate located at bits 7-0.

An ARM data processing instruction has the hexadecimal value `0xE3A0204B`. Decode the destination register and immediate value:

(a) Destination register Rd (e.g., R0-R15): _____

(b) Immediate constant (in decimal): _____

## Question 8

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** Immediate opcode decoding: opcode `1101` corresponds to `MOV`; convert immediate hex byte to decimal.

An ARM instruction has the 32-bit hexadecimal value `0xE3A03078`. What assembly instruction does this represent?

- a. `ADD R3, R0, #120`
- b. `MOV R3, #120`
- c. `SUB R3, R0, #120`
- d. `CMP R3, #120`

## Question 9

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** ARM condition code field (bits 31-28): condition `0000` denotes Equal (EQ) condition execution.

An ARM branch instruction begins with the top condition bits `0000` (hex `0x0A000012`). Which conditional branch does this encode?

- a. `BNE` (Branch if Not Equal)
- b. `BEQ` (Branch if Equal)
- c. `BGT` (Branch if Greater Than)
- d. `B` (Unconditional Branch)

## Question 10

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** ARM condition code testing: condition `0001` (NE - Not Equal) relies strictly on checking the state of the CPSR Zero (Z) flag.

If an ARM branch instruction has condition field bits `0001` (`0x1A000005`), what CPSR flag status must be met for the branch to be taken?

- a. Z = 1 (Zero flag set)
- b. Z = 0 (Zero flag clear - Not Equal)
- c. C = 1 (Carry flag set)
- d. N = 1 (Negative flag set)

## Question 11

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** Program arithmetic evaluation: integer multiplication followed by conversion from decimal to standard two-digit base-16 hexadecimal format.

An assembly program finishes executing a multiplication routine calculating $24 \times 11$. The resulting value is stored in register R0.

(a) Result in decimal: _____

(b) Result in hexadecimal (0x...): _____

## Question 12

Mark 1.00 out of 1.00

[FILL IN THE BLANK - CALCULATION]

**Technique / Concept Hint:** Multi-step arithmetic evaluation: evaluate expressions containing addition and multiplication, then express in base-16 hexadecimal.

An ARM sequence calculates the expression $(18 + 14) \times 8$. The computed value is stored in register R1.

(a) Result in decimal: _____

(b) Result in hexadecimal (0x...): _____

### PART 2: 5-STAGE RISC PIPELINE & INSTRUCTION DATAPATHS

## Question 13

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** Classic RISC execution flow: chronological progression of instructions through the 5 standard pipeline hardware segments.

What is the correct sequential execution order of stages for an instruction traveling through a classic 5-stage RISC pipeline?

- ◯ a. IF → EX → ID → MEM → WB
- ◯ b. IF → ID → EX → MEM → WB
- ◯ c. ID → IF → EX → MEM → WB
- ◯ d. IF → ID → MEM → EX → WB

## Question 14

Mark 1.00 out of 1.00

[SELECT ALL THAT APPLY]

**Technique / Concept Hint:** Pipelining performance characteristics: instruction throughput, latency vs. throughput trade-offs, and ideal speedup limits.

Which of the following statements regarding the operation of a 5-stage RISC pipeline are **TRUE**? (Select all that apply.)

- ☐ a. Multiple instructions are executed concurrently in different pipeline stages during each clock cycle.
- ☐ b. Under ideal conditions with no stalls, the throughput approaches 1 instruction completed per clock cycle.
- ☐ c. Pipelining reduces the total latency (execution time) required for a single individual instruction from start to finish.
- ☐ d. Pipeline hazards (data, control, and structural) can cause pipeline bubbles or stalls that decrease actual throughput.

## Question 15

Mark 1.00 out of 1.00

[SELECT ALL THAT APPLY]

**Technique / Concept Hint:** Functional partition of pipeline stages: verify correct mapping between stage names and underlying active datapath hardware components.

Which of the following pipeline stages are **CORRECTLY** matched with their primary functions? (Select all that apply.)

- ☐ a. **IF (Instruction Fetch):** Reads the instruction word from memory using the Program Counter (PC) address.
- ☐ b. **ID (Instruction Decode):** Reads source registers from the register file and sign-extends the immediate field.
- ☐ c. **EX (Execute):** The ALU performs arithmetic operations, logic evaluations, or computes effective memory addresses.
- ☐ d. **MEM (Memory Access):** Writes ALU results back into the destination register file for R-type instructions.

## Question 16

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** Datapath register write mechanics: identify the exact pipeline stage where register state updates are committed.

In a classic 5-stage MIPS pipeline, which stage is responsible for committing the final computed ALU value or loaded memory data back into the destination register?

- a. EX (Execute)
- b. MEM (Memory Access)
- c. WB (Write Back)
- d. ID (Instruction Decode)

## Question 17

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** Pipeline register labeling conventions: registers are named based on the two adjacent pipeline stages they decouple.

Which pipeline register holds intermediate results and control signals between the **Execute (EX)** stage and the **Memory Access (MEM)** stage?

- a. MEM/WB
- b. ID/EX
- c. EX/MEM
- d. IF/ID

## Question 18

Mark 1.00 out of 1.00

[SELECT ALL THAT APPLY]

**Technique / Concept Hint:** Pipeline register contents and timing: role of edge-triggered state storage between asynchronous stage logic.

Which of the following statements regarding pipeline registers (`IF/ID`, `ID/EX`, `EX/MEM`, `MEM/WB`) are **TRUE**? (Select all that apply.)

- [ ] a. They store data and control signals to prevent signals belonging to different instructions from interfering with one another.
- [ ] b. The `ID/EX` register holds the register values read from the register file and the sign-extended 32-bit immediate value.
- [ ] c. The `MEM/WB` register holds the data read from memory (for `lw`) or the ALU result (for arithmetic instructions).
- [ ] d. Pipeline registers are visible to the programmer and can be directly read using standard MIPS assembly instructions.

## Question 19

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** MIPS instruction formats: distinguishing R-type (three registers, opcode 0) from I-type (two registers + 16-bit constant/offset).

Which of the following MIPS instructions is an **I-type (Immediate-type)** instruction?

- a. `sub $s1, $s2, $s3`
- b. `sll $t0, $t1, 4`
- c. `lw $t0, 8($s1)`
- d. `or $t2, $t0, $t1`

## Question 20

Mark 1.00 out of 1.00

[SELECT ALL THAT APPLY]

**Technique / Concept Hint:** Structural layout comparison between MIPS instruction formats: bit allocations across opcode, registers, shamt, funct, and immediate fields.

Which of the following statements correctly compare MIPS **R-type** and **I-type** instruction formats? (Select all that apply.)

- [ ] a. R-type instructions dedicate bits 15-11 to designate the destination register `rd` .
- [ ] b. I-type instructions replace the `rd` , `shamt` , and `funct` fields with a single 16-bit immediate field (bits 15-0).
- [ ] c. Both R-type and I-type instructions contain a 6-bit opcode field located at bits 31-26.
- [ ] d. Standard R-type arithmetic instructions use an opcode of `000000` and determine the ALU operation via the `funct` field.

## Question 21

Mark 1.00 out of 1.00

[FILL IN THE BLANK - MACHINE DECODING]

**Technique / Concept Hint:** ARM data processing bit-field layout: Condition (31-28), Opcode (24-21), Rn (19-16), Rd (15-12), and Operand2 Rm (3-0).

The ARM instruction `0xE0821003` has the binary representation:

```
1110 0000 1000 0010 0001 0000 0000 0011
```

Fill in the decoded values for each field (use names like AL, ADD, or registers R0-R15):

- Condition Code (bits 31-28): _____
- Operation / Opcode (bits 24-21): _____
- Destination Register Rd (bits 15-12): _____
- First Source Register Rn (bits 19-16): _____
- Second Source Register Rm (bits 3-0): _____

## Question 22

Mark 1.00 out of 1.00

[FILL IN THE BLANK - MACHINE DECODING]

**Technique / Concept Hint:** ARM bit-field decoding: Opcode `0000` denotes bitwise AND; map nibbles to destination and source register designators.

The ARM instruction `0xE0042005` has the binary representation:

```
1110 0000 0000 0100 0010 0000 0000 0101
```

Fill in the decoded values for each field:

- Condition Code (bits 31-28): _____
- Operation / Opcode (bits 24-21): _____
- Destination Register Rd (bits 15-12): _____
- First Source Register Rn (bits 19-16): _____
- Second Source Register Rm (bits 3-0): _____

## Question 23

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** ARM instruction disassembly: decode hex nibbles into mnemonic syntax $OP \; Rd, Rn, Rm$ (opcode `0010` = SUB).

An ARM data processing instruction has the hexadecimal value `0xE0421003`. What assembly instruction does this hex code execute?

- a. `ADD R1, R2, R3`
- b. `SUB R1, R2, R3`
- c. `SUB R2, R1, R3`
- d. `MOV R1, R2`

## Question 24

Mark 1.00 out of 1.00

[MULTIPLE CHOICE - SELECT ONE]

**Technique / Concept Hint:** ARM assembly disassembly: Opcode `0100` corresponds to ADD; map Rn from bits 19-16, Rd from bits 15-12, and Rm from bits 3-0.

An ARM data processing instruction has the hexadecimal value `0xE0853006`. What assembly instruction does this hex code execute?

- a. `ADD R3, R5, R6`
- b. `ADD R5, R3, R6`
- c. `SUB R3, R5, R6`
- d. `MOV R3, R5, R6`

# Instructor Answer Key & Detailed Solutions

**Maximum Marks: 24.00**

Quiz #3 (Unit 3 - Extended) - Marking Guide & Reference Rationales

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### Question 1 (1.00 Mark) — Multiple Choice

**Correct Answer: c. To clear register R4 (set its stored value to 0)**

*Explanation:* Loading immediate 0 into a register overwrites its existing contents with zero, effectively clearing the register[cite: 8].

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### Question 2 (1.00 Mark) — Multiple Choice

**Correct Answer: a. EOR R2, R2, R2**

*Explanation:* The Exclusive-OR operation between identical values evaluates to 0 ($X \oplus X = 0$). This clears all 32 bits without requiring an immediate field.

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### Question 3 (1.00 Mark) — Fill in the Blank

**Correct Answer: 42**

*Explanation:* The instruction executes $R3 = R3 + R2$[cite: 8]. Substituting values: $28 + 14 = 42$.

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### Question 4 (1.00 Mark) — Fill in the Blank

**Correct Answer: 40**

*Explanation:* `MOV R1, #15` loads 15 into R1. Then `ADD R2, R1, #25` computes $R2 = 15 + 25 = 40$.

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### Question 5 (1.00 Mark) — Fill in the Blank

**Correct Answer: 27**

*Explanation:* The instruction computes $R6 = R6 - 18$[cite: 8]. Substituting values: $45 - 18 = 27$.

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### Question 6 (1.00 Mark) — Fill in the Blank

**Correct Answer: 28**

*Explanation:* Reverse Subtract evaluates $Rd = \text{Operand2} - Rn$. Here: $R0 = 40 - R2 = 40 - 12 = 28$.

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### Question 7 (1.00 Mark) — Fill in the Blank

**Correct Answer: (a) R2 | (b) 75**

*Step-by-step decode of 0xE3A0204B:*

Bits 15-12 = $\texttt{0010}_2$ → Destination register is **R2**[cite: 8].

Bits 7-0 = `0x4B` = $4 \times 16 + 11 =$ **75**[cite: 8].

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### Question 8 (1.00 Mark) — Multiple Choice

**Correct Answer: b. MOV R3, #120**

*Explanation:* In `0xE3A03078`, opcode bits 24-21 are `1101` (MOV), destination bits 15-12 are `0011` (R3), and immediate bits 7-0 are `0x78` ($7 \times 16 + 8 = 120$)[cite: 8].

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### Question 9 (1.00 Mark) — Multiple Choice

**Correct Answer: b. BEQ (Branch if Equal)**

*Explanation:* In ARM, condition field `0000` corresponds to EQ (Equal)[cite: 7, 8].

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### Question 10 (1.00 Mark) — Multiple Choice

**Correct Answer: b. Z = 0 (Zero flag clear - Not Equal)**

*Explanation:* Condition `0001` represents NE (Not Equal). The processor branches only when the Zero flag is 0 ($Z = 0$).

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### Question 11 (1.00 Mark) — Fill in the Blank

**Correct Answer: (a) 264 | (b) 0x108 (or 108)**

*Arithmetic:* $24 \times 11 = 264$. In hexadecimal: $264 = 16 \times 16 + 8$ → `0x108`[cite: 8].

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### Question 12 (1.00 Mark) — Fill in the Blank

**Correct Answer: (a) 256 | (b) 0x100 (or 100)**

*Arithmetic:* $(18 + 14) \times 8 = 32 \times 8 = 256$. In hexadecimal: $256 = 16^2$ → `0x100`[cite: 8].

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### Question 13 (1.00 Mark) — Multiple Choice

**Correct Answer: b. IF → ID → EX → MEM → WB**

*Explanation:* The canonical 5 stages execute chronologically: Instruction Fetch, Instruction Decode, Execute, Memory Access, and Write Back[cite: 7].

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### Question 14 (1.00 Mark) — Select All That Apply

**Correct Answers: a, b, d**

*Explanation:* Pipelining increases instruction throughput by overlapping operations across different stages (**a, b**), but hazards introduce stalls (**d**). Pipelining does not reduce individual instruction latency (**c is FALSE**).

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### Question 15 (1.00 Mark) — Select All That Apply

**Correct Answers: a, b, c**

*Explanation:* Statements **a**, **b**, and **c** are accurate[cite: 7]. Statement **d is FALSE** because the MEM stage interacts exclusively with data memory; writing back to registers occurs during the WB stage[cite: 7].

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### Question 16 (1.00 Mark) — Multiple Choice

**Correct Answer: c. WB (Write Back)**

*Explanation:* The Write Back (WB) stage updates the destination register inside the register file with either the ALU output or loaded memory data[cite: 6, 7].

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### Question 17 (1.00 Mark) — Multiple Choice

**Correct Answer: c. EX/MEM**

*Explanation:* The EX/MEM pipeline register buffers ALU calculations and control flags passing from the Execute stage into Memory Access[cite: 7].

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### Question 18 (1.00 Mark) — Select All That Apply

**Correct Answers: a, b, c**

*Explanation:* Pipeline registers insulate intermediate signals across stages (**a**), buffer decoded registers/immediates (**b**), and buffer memory/ALU outputs (**c**)[cite: 7]. They are internal hardware buffers not addressable by software (**d is FALSE**).

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### Question 19 (1.00 Mark) — Multiple Choice

**Correct Answer: c. lw $t0, 8($s1)**

*Explanation:* `lw` is an I-type instruction requiring an immediate memory offset (here, 8)[cite: 7]. The other instructions (`sub`, `sll`, `or`) are R-type[cite: 7].

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### Question 20 (1.00 Mark) — Select All That Apply

**Correct Answers: a, b, c, d**

*Explanation:* All statements are true: R-type uses bits 15-11 for `rd` and bits 5-0 for `funct`, while I-type allocates bits 15-0 to a 16-bit immediate field[cite: 7, 9]. Both share a 6-bit opcode field at bits 31-26[cite: 9].

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### Question 21 (1.00 Mark) — Fill in the Blank

**Correct Answers: AL (or Always) | ADD (or 0100) | R1 | R2 | R3**

*Bitfield Mapping of 0xE0821003:*

- Bits 31-28 (`1110`): **AL** (Always execute)[cite: 7]
- Bits 24-21 (`0100`): **ADD**[cite: 7]
- Bits 19-16 (`0010`): **R2** (Rn)[cite: 7]
- Bits 15-12 (`0001`): **R1** (Rd)[cite: 7]
- Bits 3-0 (`0011`): **R3** (Rm)[cite: 7]

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### Question 22 (1.00 Mark) — Fill in the Blank

**Correct Answers: AL (or Always) | AND (or 0000) | R2 | R4 | R5**

*Bitfield Mapping of 0xE0042005:*

- Bits 31-28 (`1110`): **AL**
- Bits 24-21 (`0000`): **AND**
- Bits 19-16 (`0100`): **R4** (Rn)
- Bits 15-12 (`0010`): **R2** (Rd)
- Bits 3-0 (`0101`): **R5** (Rm)

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### Question 23 (1.00 Mark) — Multiple Choice

**Correct Answer: b. SUB R1, R2, R3**

*Explanation:* Hex `0xE0421003` decodes to: Opcode `0010` (SUB), Rn `0010` (R2), Rd `0001` (R1), and Rm `0011` (R3) → `SUB R1, R2, R3`[cite: 7].

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### Question 24 (1.00 Mark) — Multiple Choice

**Correct Answer: a. ADD R3, R5, R6**

*Explanation:* Hex `0xE0853006` decodes to: Opcode `0100` (ADD), Rn `0101` (R5), Rd `0011` (R3), and Rm `0110` (R6) → `ADD R3, R5, R6`.

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